

1. Perceptron (artificial neuron):

A. perceptron is type of artificial neuron.

B. Input: $x_i \{i | 1, 2, \dots, n\}$

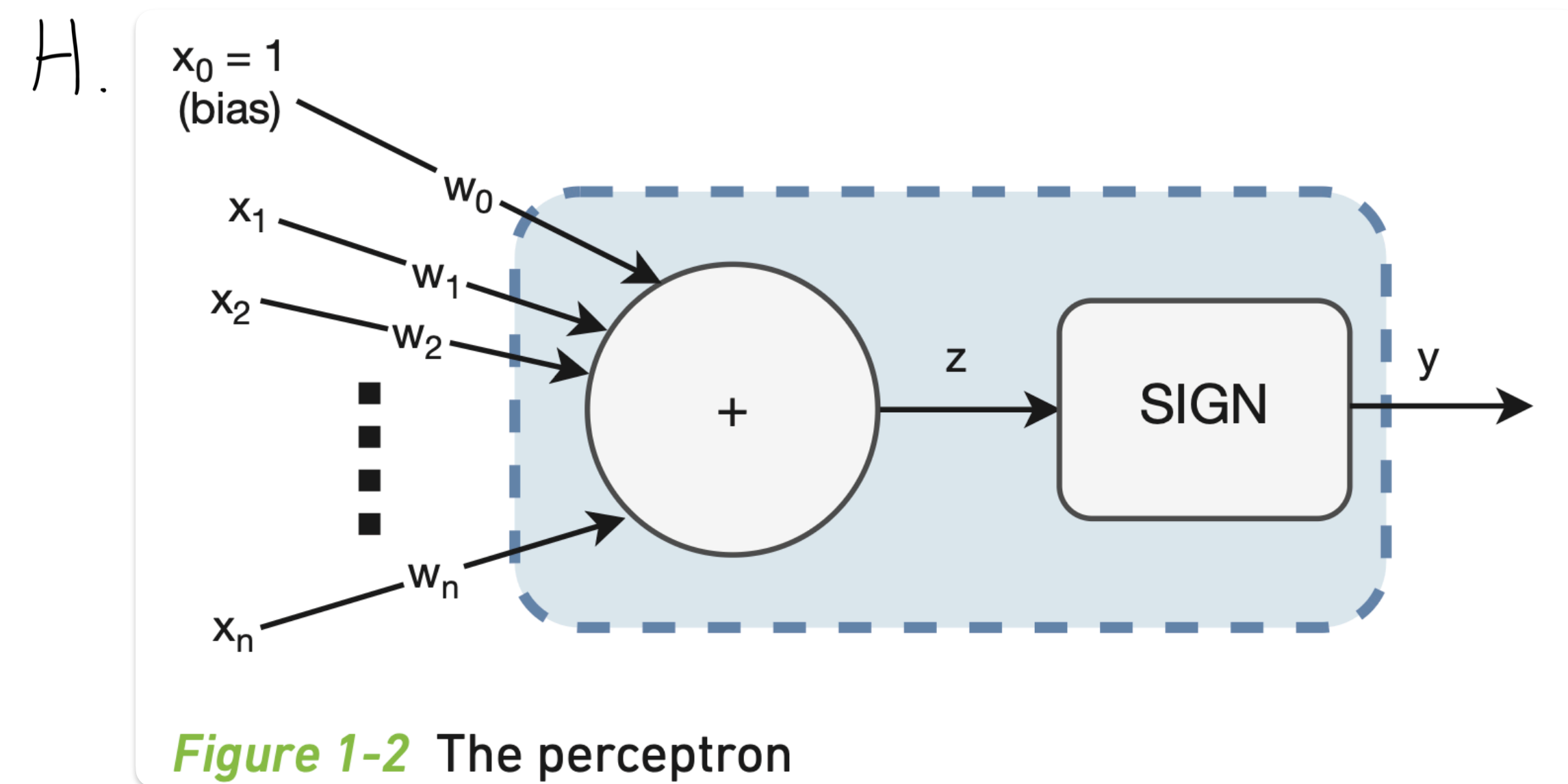
C. Bias Input: $x_0 = 1$ (always)

D. Weight: $w_i \{i | 0, 1, \dots, n\}$

E. Computational Unit: $z = \sum_0^n w_i \cdot x_i$ + $f(z) = \begin{cases} -1, & z < 0 \\ 1, & z \geq 0 \end{cases}$
(activation function)

F. Output: $y = f(z)$

G. sign function is a specific activation function used by perceptron



2. Neural Network (combined with few artificial neuron):

A. input & output of a perceptron can be "a range of real number."

B. each perceptron can implement more complex function than NAND gate, AND gate due to A.

C. use "learning algorithm" & "observed example", weight of each perceptron can be adjust to design a more accurate neural network automatically.

D. the rebuilt neuron network tends to generalize that can make sense for "not-yet-observed example" and get correct outputs.